

Radar Search Pipeline TO BE 0.7.6.3.6.5.1

Review design for a planned Radar search pipeline change

Status: TO BE

Slice: 0.7.6.3.6.5.1

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Purpose

The Radar pipeline must stop treating malformed structured LLM output as a role-specific accident. Planner, extraction, repair, revision, signal, and diagnostic probe calls all depend on machine-readable JSON. A non-JSON or schema-invalid response must follow one shared recovery contract.

Target Behavior

Every pipeline-critical structured LLM role follows this order:

1

Action	Call primary role model
Budget	OpenRouter role + total budget
Terminal diagnostic if blocked	provider unavailable or HTTP error

2

Action	Validate JSON and application schema
Budget	no extra external call
Terminal diagnostic if blocked	primary_non_json or primary_schema_invalid

3

Action	Retry primary model with strict repair context
Budget	provider retry + OpenRouter budget
Terminal diagnostic if blocked	primary_retry_non_json or primary_retry_schema_invalid

4

Action	Retry role backup model when configured
Budget	provider retry + OpenRouter budget
Terminal diagnostic if blocked	backup_non_json or backup_schema_invalid

5

Action	Stop affected branch if still invalid
Budget	no external call
Terminal diagnostic if blocked	exact recovery-exhausted reason

The planner now participates in the same contract. Extraction already has primary retry and backup recovery; this slice makes planner behavior explicit and governed by the same ADR.

Model Roles

Planner

Primary setting	OPENROUTER_PLANNER_MODEL
Backup setting	OPENROUTER_PLANNER_BACKUP_MODEL, then OPENROUTER_BACKUP_MODEL
Temperature setting	OPENROUTER_PLANNER_TEMPERATURE
Default	0

Extraction

Primary setting	OPENROUTER_EXTRACTOR_MODEL
Backup setting	OPENROUTER_EXTRACTION_BACKUP_MODEL, then OPENROUTER_BACKUP_MODEL
Temperature setting	OPENROUTER_EXTRACTOR_TEMPERATURE
Default	0

Signal/default

Primary setting	OPENROUTER_MODEL
Backup setting	future role-specific backup
Temperature setting	OPENROUTER_SIGNAL_TEMPERATURE
Default	0

Backup attempt

Primary setting	role backup model
Backup setting	none
Temperature setting	OPENROUTER_BACKUP_TEMPERATURE
Default	role temperature

Diagnostics

Technical trace and dossier metadata should show:

- role;
- primary model;
- backup model when attempted;
- attempt role: primary, primary_retry, backup;
- attempt index;
- temperature;
- budget decision;
- precise failure reason.

The payload must remain product-safe: no headers, API keys, hidden reasoning, or raw provider dumps.

Tests Required

The slice is not complete unless tests prove:

- planner non-JSON primary and primary retry can recover through backup;
- planner schema-invalid JSON triggers primary retry before backup;
- extraction non-JSON recovery still routes primary retry and backup;
- runtime config exposes role models and role temperatures;
- ADR exists and states that every new structured LLM role needs non-JSON and schema-invalid tests.

Acceptance

After implementation, run two bounded Docker smoke/evaluation passes:

1. balanced model preset;
2. light model preset.

Compare:

- terminal run outcome;
- planner/extraction attempts;
- JSON/schema failure count;
- backup usage;
- OpenRouter budget usage;
- strict recall, review recall, and precision;
- whether benchmark remains blocked or can move to broader live evaluation.